KAI-HYDRO TECHNICAL REPORT KH-TR-2026-01 · VERSION 1.0 · 30 OCTOBER 2026

Flood triggers from global data, tested on 14 river gauges: river flow or rainfall? The Guadalquivir reference case

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Companion to the research paper of the same data · interactive case: https://kai-water.charkaks0.workers.dev/case-guadalquivir

Summary for insurers, brokers and risk pools

- **The question.** Where a river has no gauge, can a global flood model carry a parametric flood trigger?
- **The test.** 14 gauges in Spain's Guadalquivir basin, 286 gauge-years (1985–2022). Triggers were set from the GloFAS global reanalysis as if no gauge existed, and scored against the floods the gauges measured.
- **Magnitude.** The model's 10-year flood is 22% of the gauged one at the median gauge. A trigger written in m³/s from it caught 6 of 60 1-in-5 flood years: no better than chance.
- **Frequency.** Written as the model's own 1-in-5 level, the same trigger caught 30 of 60 flood years, with clear skill (Peirce 0.38).
- **Four ways to set the level.** A quantile, a return period or a share of the mean flood, each from the site's own record, gave the same skill. One level for all sites paid in 0% to 78% of years from one gauge to the next.
- **River flow or rainfall.** Storm rainfall (wettest 3 days) caught 23 of 60 flood years; wet-season rainfall (October–April) caught 31, as many as river flow. Requiring both to pass cut the share of false payouts from 47% to 29%.
- **Basis risk.** 47% of that trigger's payouts came in years without a flood. Where the model ranks a site's big years well, the trigger works; where it does not, it fails. The rank correlation against any short local record predicts which (ρ = 0.80 across gauges).
- **So.** Write triggers in the model's own terms, screen basis risk with local evidence, and calibrate before the trigger enters a contract.

1. Why this test

A parametric policy pays on an index, not on an assessed loss. Its value to the buyer depends on how often the index and the loss move together. For river floods without a gauge, the index usually comes from a global flood model such as GloFAS, which publishes daily discharge for every river reach on Earth from 1979. Before a trigger from such a model is priced, three facts are needed: how big the model thinks the floods are, whether the trigger would have paid in the years the river flooded, and how often it would have paid when it did not. This report gives those facts for one basin and sets out how KAI-Hydro uses them when it writes a trigger for a client.

2. Data and method in brief

Gauges: daily flow at every Guadalquivir gauge with at least 500 km² and 25 years of record (CEDEX, *Anuario de Aforos 2021-2022*). Model: GloFAS reanalysis 4.0 at 0.05°, the cell within 10 km whose mean flow best matches the gauge (the model's best case). Each water year's largest daily flow from both; GEV and log-Pearson III fitted by L-moments, the better one kept, with 90 % bootstrap bands. Flood years are years in which a gauge reached its own 1-in-T level; triggers pay when the model reaches either the gauge's level in m³/s (fixed flow) or the model's own 1-in-T level (return period). Rainfall: catchment means of NASA POWER daily rainfall over HydroBASINS catchments. Each trigger was also set four ways (one absolute level for all gauges, each gauge's own quantile, return period, or share of its mean annual maximum) on river flow, storm rainfall and wet-season rainfall, following Model Library's Andalucía study. Full method: the paper, section 3.

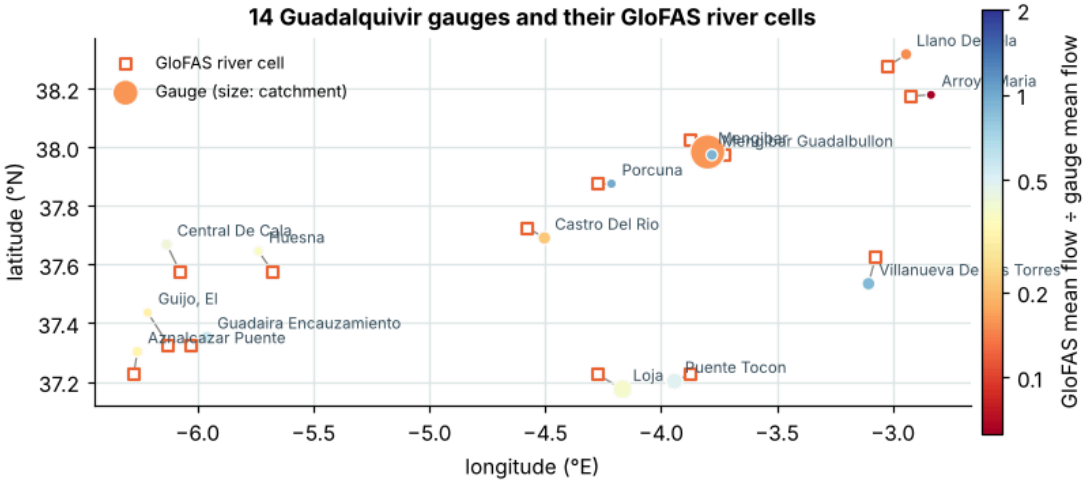


Figure 1. The 14 gauges and their GloFAS river cells. Interactive map with basemaps, rivers, admin boundaries and sub-catchments: https://kai-water.charkaks0.workers.dev/case-guadalquivir.

3. Results

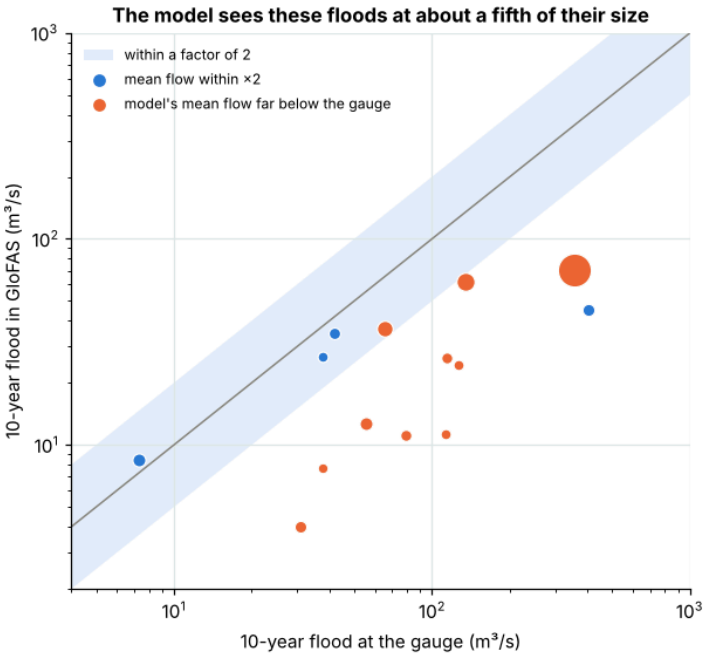


Figure 2. The 10-year flood, gauge against GloFAS, one dot per gauge.

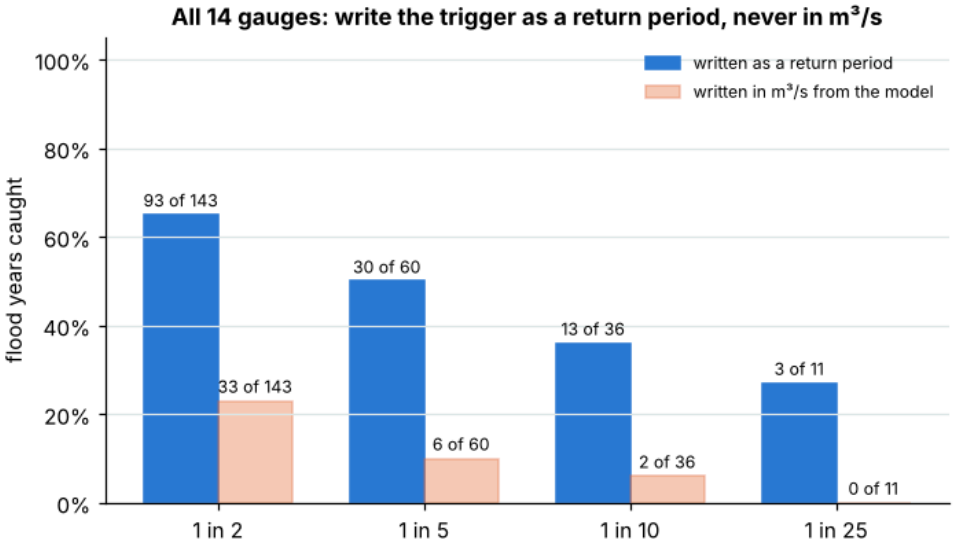


Figure 3. Share of flood years caught, pooled over all gauges, at four trigger levels, for the two ways of writing the trigger.

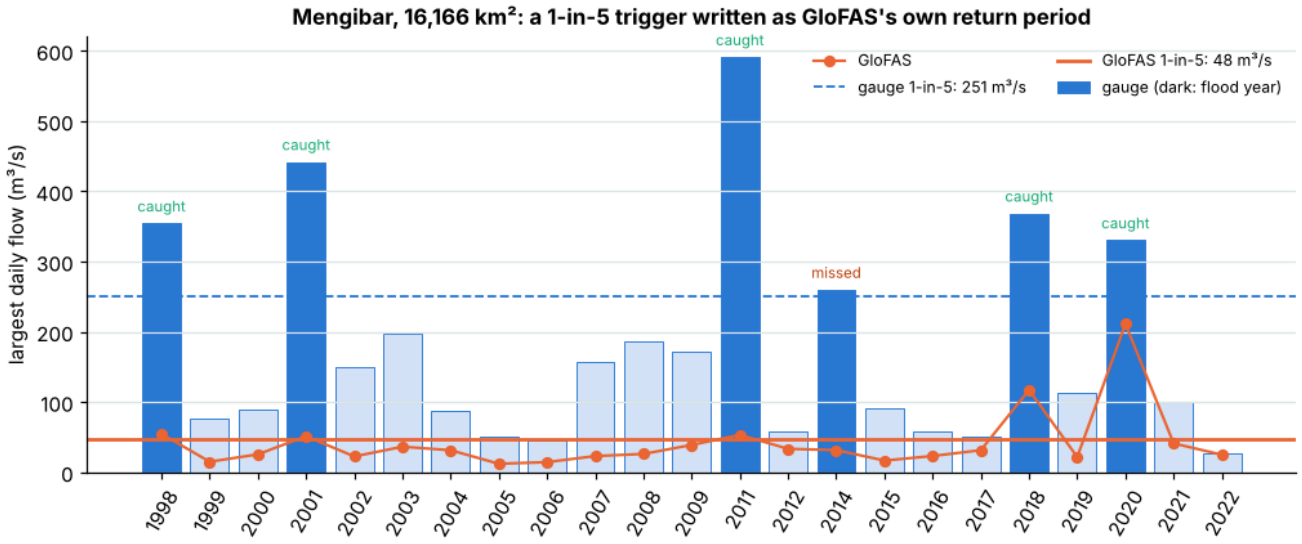


Figure 4. Year by year at Mengibar (16,166 km²): a 1-in-5 trigger written as GloFAS's own return period caught 5 of 6 flood years with no false payout; written in m³/s it would have caught none.

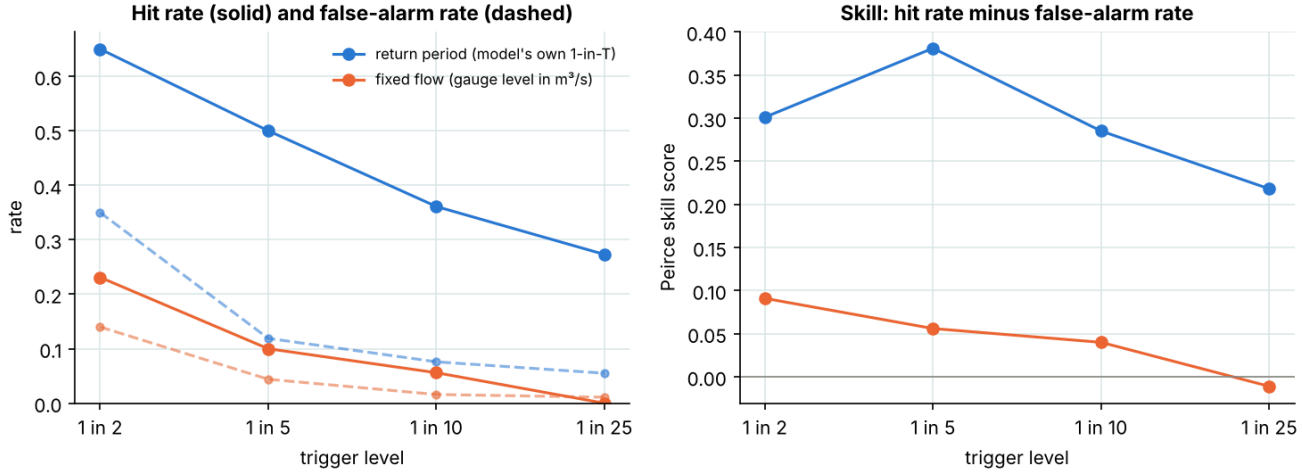


Figure 5. Hit rate, false-alarm rate and Peirce skill for both triggers at 1-in-2 to 1-in-25.

Table 1. Pooled verification, 14 gauges, 286 gauge-years.

Trigger	Written as	Caught	Missed	False payouts	Hit rate	False-alarm rate	Peirce skill	Caught by chance	p (vs chance)
1 in 2	Return period	93	50	50	0.65	0.350	0.30	71.5	2.8×10^{-7}
1 in 2	Fixed m³/s	33	110	20	0.23	0.140	0.09	26.5	0.034
1 in 5	Return period	30	30	27	0.50	0.119	0.38	12.0	1.4×10^{-9}
1 in 5	Fixed m³/s	6	54	10	0.10	0.044	0.06	3.4	0.093
1 in 10	Return period	13	23	19	0.36	0.076	0.28	4.0	1.6×10^{-5}
1 in 10	Fixed m³/s	2	34	4	0.06	0.016	0.04	0.8	0.17
1 in 25	Return period	3	8	15	0.27	0.055	0.22	0.7	0.025
1 in 25	Fixed m³/s	0	11	3	0.00	0.011	-0.01	0.1	1

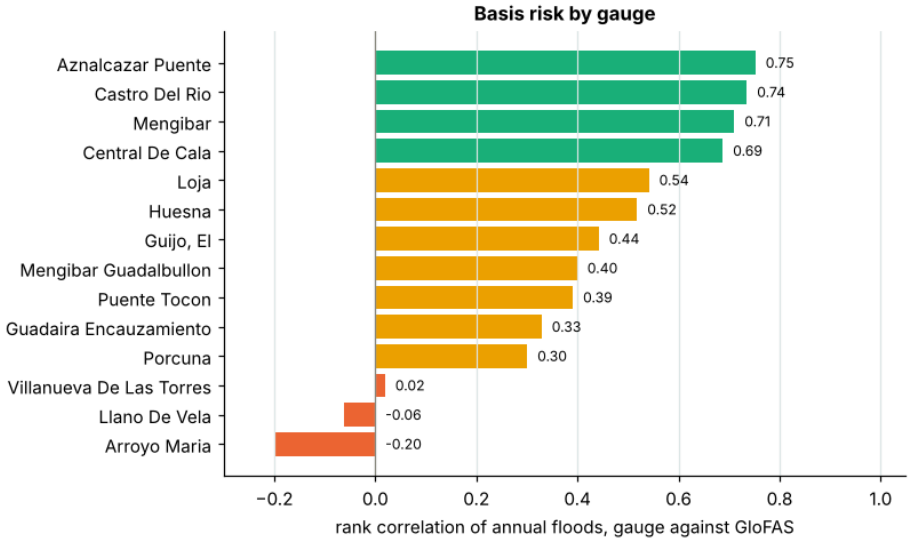


Figure 6. Basis risk by gauge: rank correlation of annual floods, gauge against GloFAS.

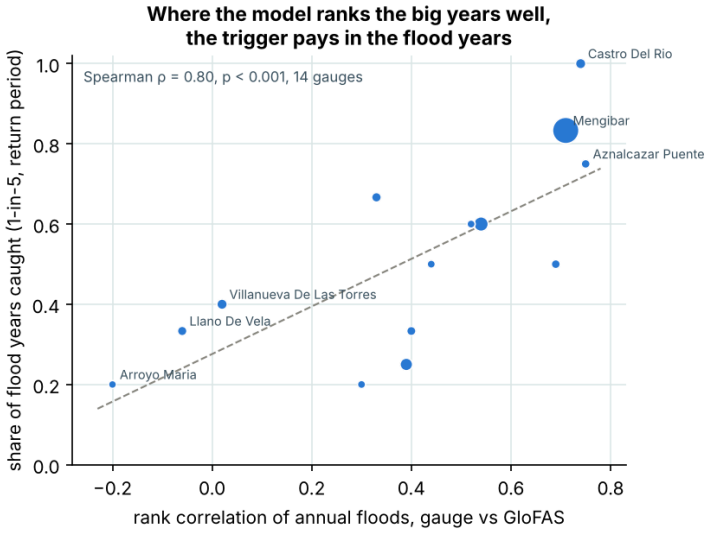


Figure 7. Share of 1-in-5 flood years caught at each gauge against the rank correlation: a screen for basis risk.

River flow or rainfall, four ways

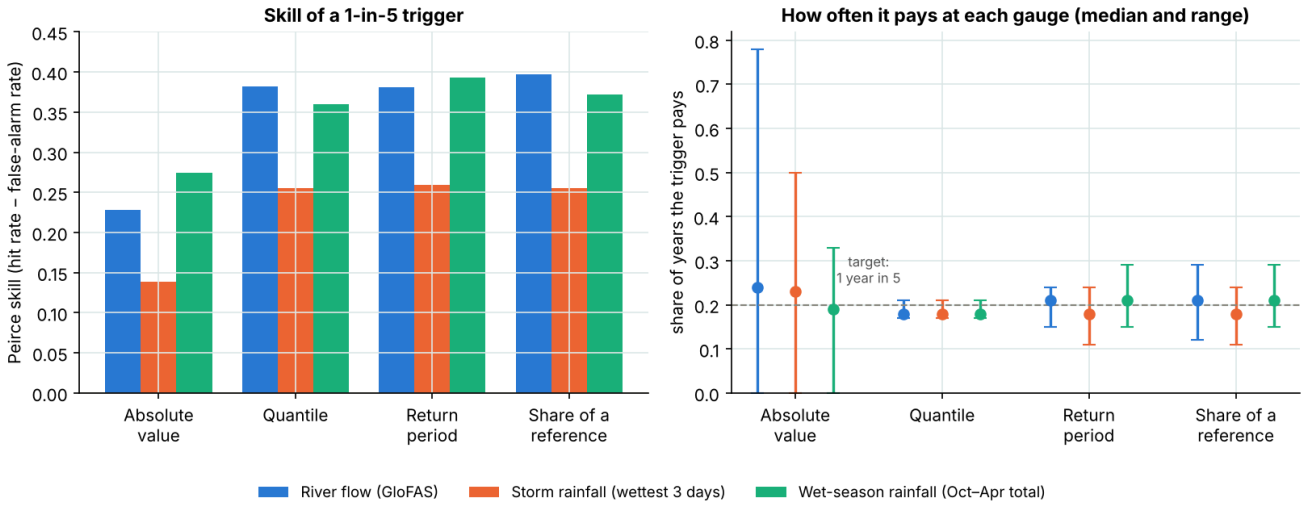


Figure 8. Triggers set to pay once in five years, four ways, on river flow and on rainfall: skill (left) and how often each pays at each gauge (right).

Table 2. 1-in-5 triggers set four ways on three indices, and two-index triggers (286 gauge-years, 60 flood years).

Index	Method	One level for all gauges	Pays, range across gauges	Caught (of 60)	False payouts	Share of payouts false	Peirce skill	p (vs chance)
River flow (GloFAS)	Absolute value	10.8 m³/s per 1,000 km²	0%–78%	28	54	66%	0.23	6.5×10^{-4}
	Quantile		17%–21%	29	23	44%	0.38	4.6×10^{-10}
	Return period		15%–24%	30	27	47%	0.38	1.4×10^{-9}
	Share of a reference	1.20 × mean	12%–29%	31	27	47%	0.40	3.4×10^{-10}
Storm rainfall (wettest 3 days)	Absolute value	64 mm	0%–50%	20	44	69%	0.14	0.019
	Quantile		17%–21%	23	29	56%	0.26	2.0×10^{-5}
	Return period		11%–24%	23	28	55%	0.26	1.3×10^{-5}
	Share of a reference	1.23 × mean	11%–24%	23	29	56%	0.26	2.0×10^{-5}
Wet-season rainfall (Oct–Apr total)	Absolute value	438 mm	0%–33%	22	21	49%	0.27	1.4×10^{-6}
	Quantile		17%–21%	28	24	46%	0.36	3.6×10^{-9}
	Return period		15%–29%	31	28	48%	0.39	6.3×10^{-10}
	Share of a reference	1.26 × mean	15%–29%	30	29	49%	0.37	4.5×10^{-9}
River and wet-season rainfall both pass		each at its own 1-in-5 level		24	10	29%	0.36	2.1×10^{-11}
Either passes		each at its own 1-in-5 level		37	45	55%	0.42	1.3×10^{-9}

4. How KAI-Hydro writes a trigger from these findings

Setting the level: what the test says

1. **From each site's own record, in the index's own terms:** its 1-in-T return period, its own quantile, or a share of its mean annual flood. All three gave the same skill here; the return period is the only one that reaches beyond the record.
2. **Never one level for all sites.** It pays at a different frequency at each site. Once each site's level is agreed, write it into the contract in m³/s or mm for settlement.
3. **River flow for event cover;** wet-season rainfall as a second index. Paying only when both agree cut the share of false payouts from about half to under a third; price the lower payout frequency.
4. **Never a flow read from another source** (a gauge study or a design report) applied to the model.

Each comes with an attachment level (payout starts) and an exhaustion level (full payout), so the payout can rise with the size of the event.

The study a client receives

1. **Snap** the site onto the river by catchment area; flag catchments too small for a 5 km model.
2. **Flood history:** 40+ years of daily discharge from GloFAS 5.0, plus any gauge or record the client holds.
3. **Return levels** with 90 % bands; levels beyond the record flagged as extrapolations.
4. **Basis-risk screen:** rank correlation of the model's big years against any local evidence (short gauge record, flood marks, satellite flood extents, claims).
5. **Back-test** of the trigger set four ways on river flow and on rainfall: hits, misses and false payouts, year by year.
6. **Calibration** where the screen says so, before the trigger enters a contract.
7. **Delivery:** a trigger sheet, a method note, and the same study as design floods for engineers.

5. Limits

One basin, strongly regulated and semi-arid; GloFAS 4.0 (the 5.0 rerun is in progress); daily means rather than instantaneous peaks; 17 to 24 years per gauge; the cell choice uses the gauge's mean flow (best case); rainfall from one product on a 50 km grid, over HydroBASINS catchments up to 2.7 times the gauge's area; flood years are defined by flow, not by losses.

Appendix A. Return levels at every gauge

Preferred fit per series (GEV or LP3, by probability-plot correlation), with 90 % percentile bootstrap bands (300 resamples).

Gauge	Return period	Gauge, m³/s	90 % band	GloFAS, m³/s	90 % band	GloFAS ÷ gauge
Mengibar	1 in 2	130.3	92–188	27.7	24–37	0.21
	1 in 5	251.1	174–339	48.0	36–71	0.19
	1 in 10	354.6	242–460	70.2	46–109	0.20
	1 in 25	520.0	350–631	114.6	56–176	0.22
	1 in 50	673.8	448–808	165.9	61–257	0.25
Loja	1 in 2	42.9	29–69	30.2	25–39	0.70
	1 in 5	91.6	58–134	47.3	38–61	0.52
	1 in 10	135.0	88–181	62.2	48–78	0.46
	1 in 25	207.1	137–247	86.1	62–105	0.42
	1 in 50	276.5	190–311	108.5	70–128	0.39
Puente Tocon	1 in 2	26.6	20–35	23.2	21–26	0.87
	1 in 5	47.3	34–63	30.7	26–35	0.65
	1 in 10	65.6	42–86	36.7	29–42	0.56
	1 in 25	94.5	52–128	45.5	32–54	0.48
	1 in 50	121.0	59–174	53.2	34–63	0.44
Villanueva De Las Torres	1 in 2	3.0	2–4	4.6	4–6	1.55
	1 in 5	5.4	4–7	6.8	5–8	1.26
	1 in 10	7.3	5–10	8.5	6–10	1.16
	1 in 25	10.2	6–15	11.1	8–12	1.09
	1 in 50	12.6	7–19	13.4	10–15	1.07
Castro Del Rio	1 in 2	21.6	18–30	3.5	3–5	0.16
	1 in 5	38.9	28–58	7.2	5–12	0.18
	1 in 10	55.5	31–84	12.7	6–25	0.23
	1 in 25	85.2	33–130	27.2	7–62	0.32
	1 in 50	115.9	33–179	49.0	8–132	0.42
Guadaira Encauzamiento	1 in 2	141.8	100–214	16.8	15–22	0.12
	1 in 5	284.4	195–411	28.9	21–45	0.10
	1 in 10	403.6	264–560	45.4	29–81	0.11
	1 in 25	589.9	331–782	84.8	38–172	0.14
	1 in 50	759.5	370–1003	137.9	45–310	0.18
Llano De Vela	1 in 2	14.6	9–21	2.0	2–2	0.14
	1 in 5	26.7	19–31	3.1	2–4	0.11
	1 in 10	30.8	24–37	4.0	3–5	0.13
	1 in 25	33.2	27–45	5.6	3–7	0.17
	1 in 50	34.0	27–53	7.2	3–10	0.21
Mengibar Guadalbullon	1 in 2	18.1	12–26	13.9	11–20	0.77
	1 in 5	32.3	23–40	24.4	18–35	0.75
	1 in 10	41.9	30–49	34.7	22–50	0.83
	1 in 25	54.1	38–62	53.5	28–80	0.99
	1 in 50	63.2	43–75	73.4	31–110	1.16
Central De Cala	1 in 2	16.6	5–40	4.0	3–5	0.24
	1 in 5	48.1	14–86	7.4	5–10	0.15
	1 in 10	78.8	26–121	11.2	7–16	0.14
	1 in 25	134.2	54–165	18.8	9–30	0.14
	1 in 50	192.0	88–217	27.5	10–50	0.14
Aznalcazar Puente	1 in 2	66.0	52–80	8.4	7–14	0.13
	1 in 5	98.4	83–114	16.6	10–31	0.17
	1 in 10	114.2	96–135	26.5	14–48	0.23
	1 in 25	129.4	103–162	48.4	24–83	0.37
	1 in 50	138.0	106–178	76.3	34–120	0.55
Huesna	1 in 2	28.0	18–53	5.9	4–11	0.21
	1 in 5	74.3	40–133	14.2	7–28	0.19
	1 in 10	126.1	59–216	24.4	10–46	0.19

Gauge	Return period	Gauge, m³/s	90 % band	GloFAS, m³/s	90 % band	GloFAS ÷ gauge
	1 in 25	232.8	94–364	47.0	18–82	0.20
	1 in 50	358.7	120–523	75.9	28–124	0.21
Guijo, El	1 in 2	47.3	36–60	7.3	7–8	0.15
	1 in 5	84.4	62–108	9.5	8–11	0.11
	1 in 10	113.3	74–146	11.2	9–13	0.10
	1 in 25	155.8	83–202	13.7	10–17	0.09
	1 in 50	192.3	87–255	15.9	11–21	0.08
Porcuna	1 in 2	9.5	5–19	6.2	4–10	0.66
	1 in 5	24.0	12–39	15.9	9–25	0.66
	1 in 10	37.6	20–54	26.8	14–46	0.71
	1 in 25	61.4	35–76	47.8	20–101	0.78
	1 in 50	85.4	52–96	70.3	23–178	0.82
Arroyo Maria	1 in 2	19.8	17–24	3.1	3–4	0.16
	1 in 5	29.4	24–37	5.2	4–7	0.18
	1 in 10	37.5	30–47	7.7	5–12	0.21
	1 in 25	50.4	38–62	13.2	6–21	0.26
	1 in 50	62.4	45–77	20.2	7–34	0.32

Appendix B. The 1-in-5 trigger at every gauge

Gauge	Catchment km²	Years	Mean-flow ratio	Q10 gauge, m³/s (90 % band)	Q10 GloFAS, m³/s (90 % band)	Q10 ratio	Rank corr.	Flood years	Caught	False payouts
Mengibar	16,166	23	0.16	355 (242–460)	70 (46–109)	0.20	0.71	6	5	0
Loja	4,210	23	0.39	135 (88–181)	62 (48–78)	0.46	0.54	5	3	2
Puente Tocon	2,944	18	0.48	66 (42–86)	37 (29–42)	0.56	0.39	4	1	3
Villanueva De Las Torres	1,685	21	0.90	7 (5–10)	9 (6–10)	1.16	0.02	5	2	3
Castro Del Rio	1,665	17	0.22	56 (31–84)	13 (6–25)	0.23	0.74	2	2	1
Guadaira Encauzamiento	1,318	18	0.56	404 (264–560)	45 (29–81)	0.11	0.33	3	2	2
Llano De Vela	1,240	20	0.15	31 (24–37)	4 (3–5)	0.13	-0.06	3	1	2
Mengibar Guadalbullon	1,111	17	0.93	42 (30–49)	35 (22–50)	0.83	0.40	3	1	2
Central De Cala	1,010	19	0.42	79 (26–121)	11 (7–16)	0.14	0.69	6	3	1
Aznalcazar Puente	960	22	0.32	114 (96–135)	26 (14–48)	0.23	0.75	4	3	2
Huesna	678	21	0.37	126 (59–216)	24 (10–46)	0.19	0.52	5	3	2
Guijo, El	672	23	0.31	113 (74–146)	11 (9–13)	0.10	0.44	4	2	2
Porcuna	670	20	0.99	38 (20–54)	27 (14–46)	0.71	0.30	5	1	2
Arroyo Maria	583	24	0.05	37 (30–47)	8 (5–12)	0.21	-0.20	5	1	3

Appendix C. Rainfall windows

Each rainfall index written as the catchment's own 1-in-5 level and scored against the same 60 flood years. The 3-day and October–April indices were chosen in advance; the others are a sensitivity test.

Index	Median rank corr. with gauge	Caught (of 60)	False payouts	Peirce skill
River flow (GloFAS), for comparison	0.42	30	27	0.38
Rainfall, wettest day	0.18	19	37	0.15
Rainfall, wettest 3 days	0.30	23	28	0.26
Rainfall, wettest 5 days	0.34	23	33	0.24
Rainfall, wettest 10 days	0.42	25	41	0.23
Rainfall, wettest 30 days	0.50	26	32	0.29
Rainfall, October–April total	0.52	31	28	0.39

Appendix D. Data package and how to reproduce

File	What it holds
data/annual_maxima.csv	286 rows: gauge_id, water_year, gauge_max_m3s, glofas_max_m3s
data/gauges.csv	14 gauges: catchment, position, record length, matched GloFAS cell, mean-flow ratio
results/return_levels.csv	1-in-2 to 1-in-50 levels with 90 % bands, both sources, per gauge
results/trigger_scores.csv	caught, missed and false payouts at 1-in-2 to 1-in-25, both triggers, pooled
results/skill_scores.csv	hit rate, false-alarm rate, Peirce skill, test against chance
results/per_gauge_1in5.csv	Table 1 of the paper
results/kai_engine_results.json	the KAI engine's published results, checked by the notebook
01_global_model_flood_triggers.ipynb	the notebook: fits, triggers, figures and the check against the engine
data/rainfall_maxima.csv	each water year's largest 1-, 3-, 5-, 10- and 30-day catchment rainfall, and the October–April total (mm)
data/rain_catchments.csv · rain_cell_weights.csv	HydroBASINS sub-catchments used for each gauge, and the NASA POWER cells with their weights
results/trigger_methods.csv · trigger_methods_sites.csv	triggers set four ways on three indices: pooled scores, and how often each pays at each gauge
results/rain_windows.csv · rain_vs_river.json	rainfall windows compared; river against wet-season rainfall, and the two combined
flood_tools.py · rebuild_data.py	shared functions (L-moments, GEV, LP3, bootstrap, scoring) and the rebuild script
rebuild_rainfall.py · compare_triggers.py	catchments and catchment rainfall; the four-way comparison and its figure
paper/paper_stats.py	the skill scores, per-gauge table, McNemar and Spearman tests, and the skill and basis-risk figures
paper/build_documents.py · print.js	builds the paper and this report from the result files, and prints them to PDF

To reproduce: install Python 3.11 or newer and the packages in *requirements.txt*, then run the notebook from its folder; it stops if any trigger count drifts from the published results. To rebuild the derived data from the sources, run *rebuild_data.py* on a connection that CEDEX serves; it fetches the CEDEX files (SHA-256 checked) and the GloFAS cells, and writes **_rebuilt.csv* files to compare with the published ones. *rebuild_rainfall.py* rebuilds the catchments and catchment rainfall from HydroBASINS (level-8 shapefile, downloaded separately) and the NASA POWER API, and *compare_triggers.py* sets and scores the triggers four ways on the three indices.

Data credits and licence

Fuente: CEDEX, Anuario de Aforos 2021-2022 (MITECO). Contains modified Copernicus Emergency Management Service information 2026 (GloFAS 4.0), via Open-Meteo (CC BY 4.0). GloFAS data are provided for information purposes only and are not a flood warning. Rainfall: NASA Langley Research Center POWER Project, funded through the NASA Earth Science/Applied Science Program. Catchments: HydroBASINS (Lehner and Grill, 2013). Four-way comparison after Model Library, *Four ways to set a parametric trigger, tested on Andalucía rainfall and yields* (9 October 2026). This report: © 2026 Charles Kakooza, CC BY 4.0.